

LUEGDE-PAENBRUCH, Germany

WMO: 104330

Lat: 51.8664N

Lon: 9.2708E

Elev: 259

StdP: 98.25

Time Zone: 1.00 (EUC)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-8.2	-13.2	1.2	-9.7	-10.9	1.5	-6.9	10.9	6.3	9.8	6.0	2.2	10	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	27.9	19.0	25.8	18.3	23.9	17.4	19.9	25.4	19.1	24.1	18.1	22.5	3.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.5	21.6	17.4	12.8	20.7	16.6	12.2	19.9	58.2	25.1	55.3	24.2	52.2	22.3	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.5	-13.4	30.7	4.1	2.7	-16.3	32.6	-18.7	34.2	-21.0	35.7	-23.9	37.7		
(5)			-13.6	21.6	4.0	1.0	-16.5	22.3	-18.8	22.9	-21.0	23.4	-23.9	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	0.5	0.7	4.0	8.8	11.8	15.1	17.1	16.8	13.5	9.2	5.2	0.8		
	DBStd	7.15	4.84	4.64	4.44	4.02	3.43	3.28	3.32	2.85	3.12	3.77	3.46	4.22		
	HDD10.0	1334	296	262	189	69	21	2	0	0	5	61	147	284		
	HDD18.3	3588	554	494	444	286	203	108	65	64	149	283	394	542		
	CDD10.0	846	0	1	3	33	78	154	220	211	108	36	3	0		
	CDD18.3	58	0	0	0	0	1	9	27	17	3	0	0	0		
	CDH23.3	525	0	0	0	2	11	74	263	152	24	1	0	0		
	CDH26.7	117	0	0	0	0	0	9	65	41	1	0	0	0		
(14) Wind	WSAvg	3.2	4.0	3.6	3.4	3.1	3.0	2.8	2.6	2.6	2.7	3.2	3.5	3.6		
(15) Precipitation	PrecAvg	894	86	69	67	52	67	72	87	82	71	73	81	88		
	PrecMax	1276	175	161	166	111	174	117	184	160	173	243	142	193		
	PrecMin	682	11	21	8	5	23	36	37	33	9	21	5	35		
	PrecStd	155	41	35	39	24	37	21	39	36	39	43	37	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	12.4	16.5	23.1	24.9	27.9	31.0	32.2	26.0	20.3	14.0	10.9		
		MCWB	9.7	8.5	10.4	13.9	18.0	19.6	19.0	19.0	18.2	16.2	11.6	9.4		
	2%	DB	9.9	10.0	14.2	20.5	22.3	25.5	28.6	27.5	23.3	17.6	12.2	8.7		
		MCWB	8.7	8.2	9.2	13.0	15.6	18.8	19.2	18.5	17.0	14.3	10.5	7.3		
	5%	DB	8.4	8.3	12.4	18.2	20.4	23.4	26.3	24.8	21.1	15.8	11.1	7.6		
		MCWB	7.2	6.8	8.7	11.8	14.1	17.4	18.3	17.7	16.0	13.5	9.7	6.6		
	10%	DB	6.9	6.9	10.3	16.0	18.5	21.4	24.1	22.5	19.0	14.4	9.8	6.3		
		MCWB	5.8	5.5	7.6	10.9	13.5	16.3	17.4	16.9	15.1	12.6	8.6	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.4	11.5	14.7	18.9	21.4	21.1	20.7	19.5	16.4	12.1	9.6		
		MCDB	10.9	11.1	14.8	20.8	23.0	26.3	27.4	26.6	23.3	19.5	13.4	10.9		
	2%	WB	8.8	8.3	10.4	13.5	17.1	19.4	19.9	19.6	17.6	14.9	10.8	7.7		
		MCDB	9.8	9.8	13.0	19.0	20.8	24.4	25.7	24.9	22.0	16.8	12.0	8.3		
	5%	WB	7.3	7.0	9.1	12.5	15.5	18.2	19.0	18.6	16.6	13.8	9.8	6.6		
		MCDB	8.2	8.2	11.7	17.3	19.1	22.1	24.4	23.2	20.4	15.5	11.0	7.6		
	10%	WB	5.9	5.5	7.7	11.4	14.1	17.0	18.1	17.6	15.5	12.7	8.7	5.3		
		MCDB	6.9	6.8	10.2	15.4	17.2	20.3	22.7	21.5	18.5	14.2	9.8	6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	6.6	8.7	8.7	8.9	9.2	8.6	8.1	6.2	4.1	3.8		
		MCDBR	4.7	5.4	10.1	12.8	12.1	12.1	13.7	13.0	11.7	7.8	4.9	4.3		
	5% WB	MCWBR	4.4	4.1	6.3	6.8	6.6	6.2	5.5	5.9	6.2	5.2	3.7	3.8		
		MCWBR	4.5	5.0	8.5	11.0	10.6	10.8	11.8	10.4	10.2	7.3	4.6	4.3		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.332	0.367	0.403	0.406	0.405	0.418	0.412	0.379	0.349	0.330	0.306		
		taud	2.391	2.364	2.288	2.229	2.251	2.273	2.243	2.278	2.364	2.431	2.408	2.378		
	Ebn at Noon		699	784	819	831	847	850	830	813	800	754	664	634		
		Edh at Noon	60	81	105	125	129	127	129	118	97	75	59	52		
(44) All-Sky Solar Radiation	RadAvg		0.67	1.35	2.44	3.94	4.77	5.16	4.91	4.17	3.02	1.75	0.79	0.51		
	RadStd		0.09	0.21	0.31	0.48	0.51	0.44	0.66	0.36	0.40	0.17	0.11	0.06		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (41 neighbors)		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	-127	N/A	N/A	

Nomenclature: See separate page